INTERNET PROTOCOL

NAME	ACCESS NUMBER
NANTANDA EVE HARRIET PAULINE	B39045
KAMUGISHA JACKSON	B39090
FREDAH KANSIIME	B39123
YAWE JOHN JAYDEN	B39111
NSAAWA MUGABI JONATHAN	B40456
BABIRYE TENDO JAEL	B40149
AGWANG LETICIA WINNIEFRED	B40602
MUYOMBA ANTONIO	B41617
KASASA ALBERT BWETE	B39110
AHIMBISIBWE BOSCO	B39926

INTRODUCTION

An IP Address is a unique numeric label assigned to every device connected to a computer that uses Internet Protocol for communication.

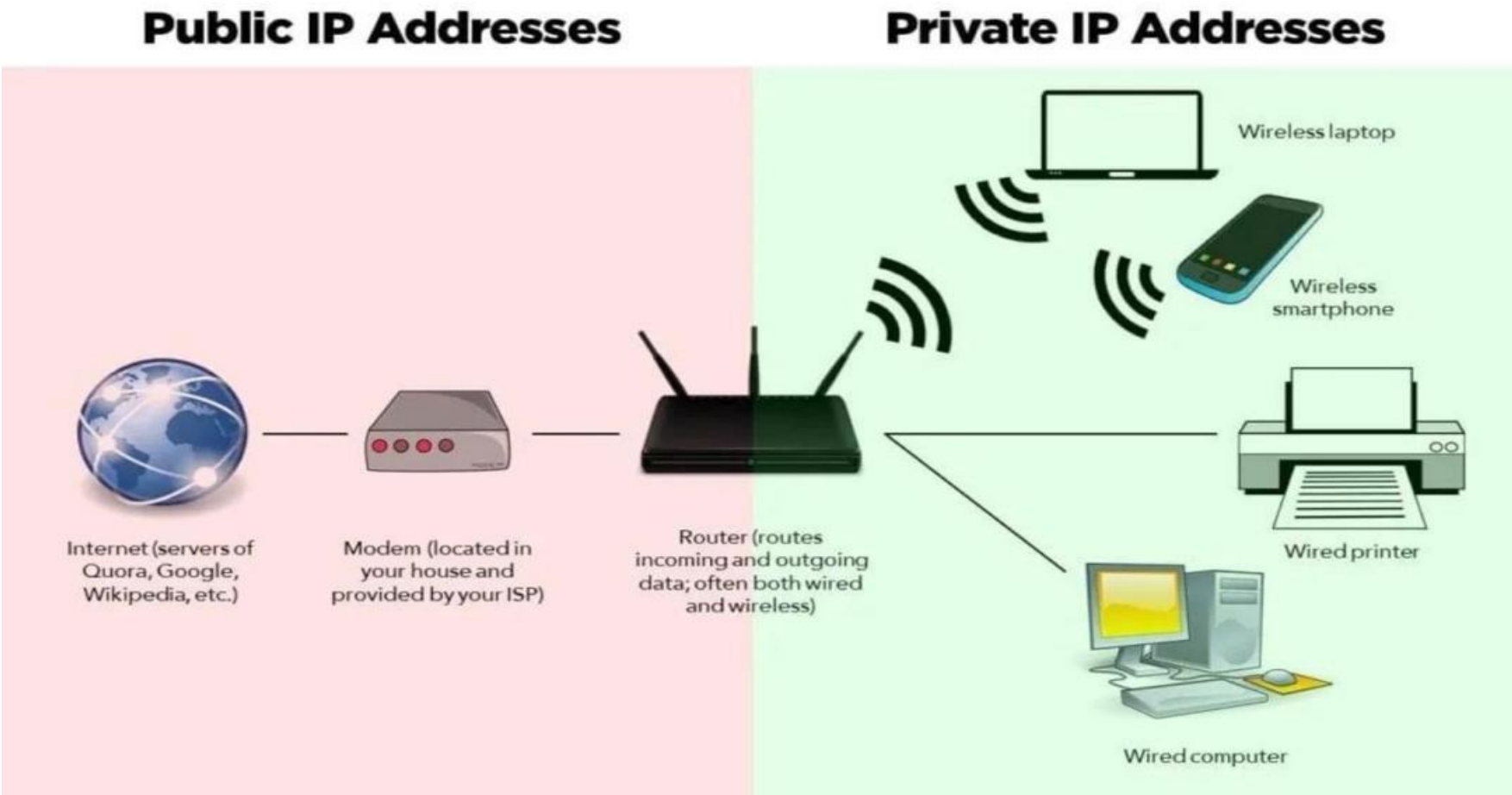
MAIN FUNCTIONS

- *Network Interface Identification*: Identifies a specific NI on a network.
- *Location Addressing*: It provides the digital location of the devices so data packets reach the correct destination.

TYPES OF IP

- Public IP & Private IP
- IPv4 & IPv6
- Static & Dynamic

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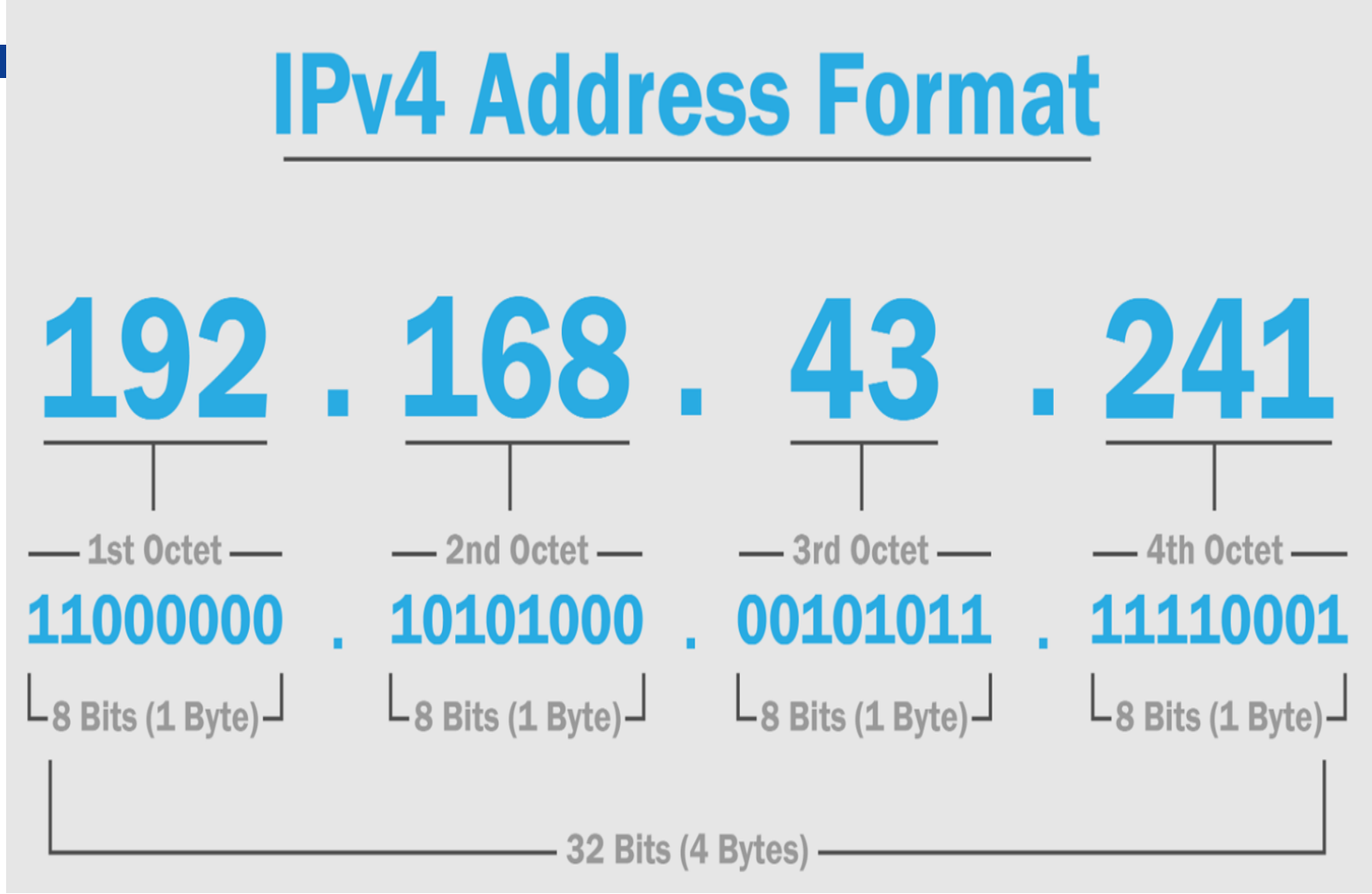


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IPv4

IPv4 is the older version of the Internet Protocol used to identify devices on a network.

- It uses a 32-bit address.
- It is usually written as four numbers separated by dots.
- Example: 192.168.1.10
- It provides about 4.3 billion addresses.



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How to Create a Private IPv4 Address

Structure and Format: Build the address using four numbers from 0 to 255 separated by dots (A.B.C.D).

Define Network and Device: Use an easy private range like 192.168.x.x, where the third number names your network (e.g., 192.168.5) and the last number names your device (2 to 254).

Pair with a Subnet Mask: Use 255.255.255.0 (or /24) to specify that the first three numbers represent the network and the last represents the device.

Validation Checklist: Verify that every number is between 0 and 255, avoid leading zeros, ensure the last number is not 0 (network) or 255 (broadcast), and check that the address is not already in use.

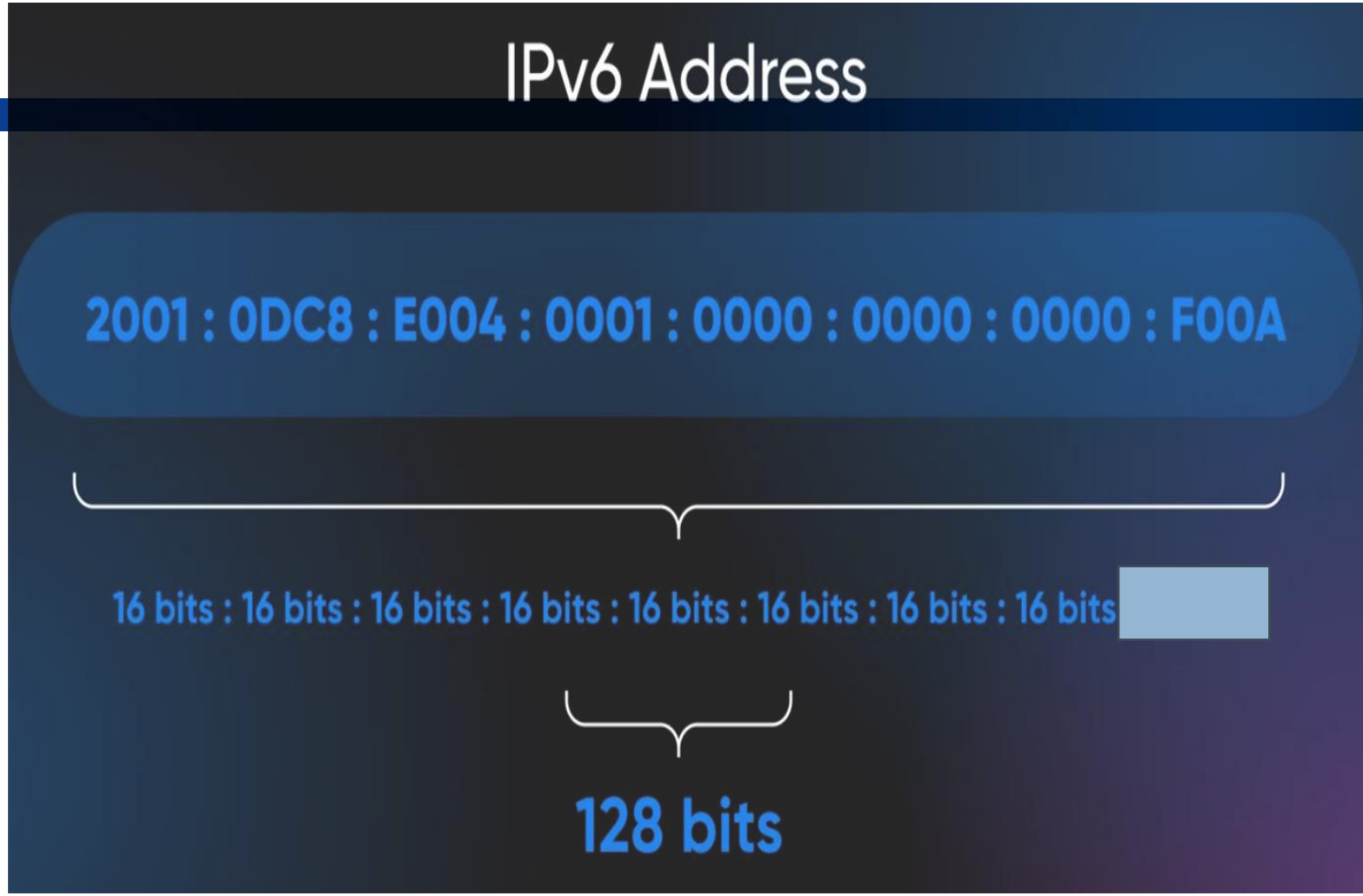
Device Setup: Enter the IP address, subnet mask, and gateway manually in your network settings, or let your router's DHCP assign it automatically.

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IPv6

IPv6 is the newer version of the Internet Protocol. It was developed mainly because IPv4 addresses are limited.

- It uses a 128-bit address.
- It is written using hexadecimal numbers.
- Example: 2001:db8:85a3::8a2e:370:7334
- It provides a very large number of addresses.



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DIFFERENCES BETWEEN IPv4 and IPv6

	IPv4	IPv6
Address length	32-bit numeric labels	128-bit alphanumeric labels
Format	Written in decimal format	Written in hexadecimal format
Capacity	Allows 4.3billion unique addresses (2^{32})	Provides an almost limitless number of unique addresses (2^{128})
Structure	4 octets separated by periods (0 to 255)	8 hextets are separated by colons (0-9 AND A-F)
Example	192.168.1.1	2001:db8:0:1234:0:567:8:1

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What is subnetting?

Subnetting splits one large IP network into smaller, separate networks called subnets.

1. **Less broadcast traffic**
   Each subnet is its own broadcast domain.

2. **Better security**
   Isolate groups such as labs, staff and servers.

3. **Efficient addressing**
   Give each network only the addresses it needs.

192.168.1.0/24
One network · 254 hosts

▼ split into ▼

Admin /26 · 62 hosts	**Labs** /26 · 62 hosts
Wi-Fi /26 · 62 hosts	**Servers** /26 · 62 hosts

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THE SUBNET MASK

The mask says which of the 32 bits identify the network. Subnetting borrows bits from the host part.

Network · 24 bits	Borrowed · 2	Host · 6 bits

192.168.1.0 /26 = mask 255.255.255.192 (26 network bits)

4

subnets = 2^n, n = 2 borrowed bits

62

usable hosts = $2^h - 2$, h = 6 host bits

64

block size = 256 − 192, so subnets start every 64

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WORKED EXAMPLE

Split 192.168.1.0/24 into four /26 subnets:

Subnet	Usable host range	Broadcast
192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.127
192.168.1.128/26	192.168.1.129 – 192.168.1.190	192.168.1.191
192.168.1.192/26	192.168.1.193 – 192.168.1.254	192.168.1.255

Quick method: block size = 256 − 192 = 64. Each subnet starts 64 addresses after the last; the first address is the network ID and the last is the broadcast.

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MAGIC NUMBER

A magic number is a number we use to quickly find the subnet ranges when doing subnetting.

The formula is:

Magic number = 256 − subnet mask value

Example:

IP: 192.168.1.70

Subnet mask: 255.255.255.192

The last number is 192.

So:

Magic number = 256 − 192 = 64

Therefore, the subnets increase by 64:

- 192.168.1.0
- 192.168.1.64
- 192.168.1.128
- 192.168.1.192

So the magic number is 64.

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Host Numbers

Reserved Addresses (.0 & .255)

- Each number in an IPv4 address consists of 8 bits, yielding 256 total values (0 to 255).
- **.0 (Network Address):** Represents the name of the network itself (e.g., 192.168.5.0).
- **.255 (Broadcast Address):** Used to simultaneously transmit messages to every device on that network.

Usable Device Ranges (.2 to .254)

- **Technical Allowance:** .1 through .254 are technically assignable (254 total addresses).
- **Gateway Convention:** .1 is conventionally reserved for the router (gateway), making .2 to .254 the safe standard for devices.
- **Small Subnets:** Smaller subnets (like /29) feature drastically reduced usable host pools.

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